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Adaptive information-maximization encoding for ghost imaging: a general Bayesian framework under experimental physical constraints

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Ghost imaging (GI) has demonstrated diverse imaging capabilities arising from its encoding-decoding-based computational imaging mechanism. Accordingly, information-theoretic studies have emerged as a promising direction for characterizing the fundamental performance limits of GI and related computational imaging paradigms. Within the information-theoretic framework, the fundamental question of information-optimal encoding design in GI remains largely unexplored, primarily due to the intractability of the prior probability density function (PDF) of an unknown scene. Here, by leveraging the ability of recursively estimating the PDF of the object to be imaged via Bayesian filtering, we establish an adaptive information-maximization encoding (AIME) design framework. Based on the adaptively estimated posterior PDF from previously acquired measurements, the expected information gain of subsequent detections is evaluated and maximized to design the corresponding encoding patterns in a closed-loop manner. Under this framework, the theoretical form of the information-optimal encoding under the total-energy-constrained regime is analytically derived. Corresponding experimental results show that, GI employing information-optimal encoding together with Bayesian-filtering-based reconstruction achieves improved imaging performance compared to conventional fixed point-to-point imaging without relying on additional heuristic regularization, particularly in low-signal-to-noise ratio regimes. Moreover, the proposed AIME strategy is compared with existing encoding strategies under a representative experimental constraint, which consistently shows that it enables significantly enhanced information acquisition capability compared with existing encoding strategies, leading to substantially improved imaging quality. These results establish a principled information-theoretic foundation for optimal encoding design in computational imaging paradigms, provided that the forward model can be accurately characterized. © 2026 Chinese Laser Press

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1. INTRODUCTION

Ghost imaging (GI) [1] has demonstrated a broad range of capabilities, including acquisition of high-dimensional (H-D) information through low-dimensional detection [2], enhancement of two-dimensional resolution via H-D light-field information [3], non-locally decoupling detection efficiency and imaging resolution [4], and reduction of the unnecessary detection redundancy [5]. These advantages have enabled its diverse applications in imaging LiDAR [6,7], single-shot H-D imaging

cameras [8,9], super-resolution microscopy [10,11], low-dose and multi-dimensional radiographic imaging [12,13], imaging in special spectral bands [14,15], and task-oriented imaging [16,17]. From the perspective of imaging information acquisition [18,19], GI can be interpreted as an encoding-decoding-based computational imaging paradigm [20], where the imaging object is encoded by randomly fluctuating light fields, and the desired image is decoded through high-order light-field correlation. This formulation naturally places GI within an information-theoretic framework, under which analyses

of information acquisition efficiency and fundamental performance are in principle attainable [21–23].

Within the information-theoretic framework, a fundamental question in GI, and more broadly in such computational imaging paradigms, concerns the existence and design of information-theoretically optimal encoding [24]. In information theory, the encoding optimality is commonly evaluated using information-content measures, such as mutual information (MI) [25] and Fisher information [26]. However, the probability density function (PDF) of the object to be imaged, which is required for calculating these information measures, is generally intractable in practical imaging systems. As a result, information-theoretic studies on encoding design of GI have so far been confined to very restrictive assumptions or special scenarios. For example, Ref. [27] considers a particular sparse point-target scene prior, whereas practical imaging scenes generally require more flexible statistical descriptions. As a practical compromise, encoding optimization has been extensively explored within the framework of compressed sensing (CS), motivated by the principle that incoherent sampling can preserve the information of sparse signals with high efficiency [28]. Assuming sparsity of the image in a specific transform domain, the encoding patterns in GI can be optimized by minimizing mutual coherence of the sensing matrix [5,29,30]. However, such strategy does not explicitly incorporate the measurement-noise statistics or sparsifying-model mismatch into the encoding objective, which limits its applicability under low-signal-to-noise ratio (SNR) and undersampled conditions [31]. Although robust CS theory [32] provides powerful tools for noisy and incomplete measurements, translating such noise robustness into a practical encoding-optimization criterion for GI is nontrivial and remains less explored. Thus, existing approaches along this line remain theoretically incomplete, and are difficult to apply to GI under realistic low-SNR conditions.

Recently, by modeling the linear computational imaging system within the Bayesian filtering framework, Du *et al.* [33] proposed a recursive approach to estimating the PDF of the imaging object from already-acquired detection signals. This formulation suggests a promising route for evaluating information-theoretic measures and designing information-optimal encoding light fields in an adaptive manner. However, from existing studies on adaptive GI [34,35], it remains unclear whether adaptive measurements can consistently yield fundamental performance gains, particularly from an information-theoretic perspective. More broadly, in the context of adaptive CS, the superiority of adaptive sensing over non-adaptive measurements has long remained theoretically inconclusive [36–38], especially in noisy and practically constrained regimes.

In this article, we explicitly investigate the possibility to adaptively establish the information-optimal encoding for GI, and propose an adaptive information-maximization encoding (AIME) framework. It enables quantitative evaluation of the information content acquired in the encoding-detection process, and facilitates the design of optimal encoding under practical constraints through information maximization. It is worth noting that, different from the previous Bayesian-filtering-based GI study in Ref. [33], where adaptive measurement

design was mainly implemented by selecting or switching among predefined encoding sets, AIME directly optimizes the encoding pattern itself after each measurement. Theoretical analysis and experimental results consistently reveal that the form of information-optimal encoding is intrinsically governed by the underlying physical constraints. Under an ideal total-energy constraint corresponding to fixed illumination energy, the information-maximization encoding converges to a point-like illumination pattern, leading to an adaptive scanning strategy whose order is determined by the evolving object statistics rather than by a predetermined sequence. This result suggests that point-wise scanning can naturally emerge as an information-optimal encoding mode under the total-energy constraint. In the experiments, the AIME framework combined with Bayesian filtering reconstruction shows improved imaging performance compared with conventional fixed point-to-point imaging modes, particularly in low-SNR regimes. Under realistic modulation constraints imposed by experimental hardware, although the optimal encoding no longer reduces to a point-like structure, it still emerges as an object-dependent adaptive pattern. In this regime, comparisons with representative encoding strategies show that AIME consistently provides improved imaging quality across a wide range of sampling ratio (SR) and SNR conditions. Further evaluation of the acquired information of different encoding strategies shows good agreement with image quality, indicating that the performance gain originates from more efficient information acquisition rather than heuristic estimation effects. The proposed framework establishes a principled information-theoretic foundation for characterizing and approaching the fundamental performance of computational imaging paradigms.

2. RESULTS

A. Principle of AIME for GI

We formulate AIME as a closed-loop information-theoretic framework for GI, in which the encoding patterns are treated as variables to be optimized according to the evolving statistical state of the estimated imaging object. As illustrated in Fig. 1(b), at each measurement step, the statistical state is represented by a posterior PDF inferred from previously detected signals, based on which the expected information gain of a candidate encoding is evaluated. The subsequent encoding is then determined by maximizing the information criterion under explicit physical constraints followed by a recursive update of the posterior PDF after measurement. Within this formulation, encoding, detection, and statistical inference in GI are integrated into a unified framework, enabling information-optimal encoding to be defined and implemented sequentially without requiring special assumptions on prior knowledge of the object statistics.

1. Bayesian Formulation for Recursive Posterior PDF Tracking

To enable information-theoretic evaluation and optimization of encoding, the statistical state of the imaging object is characterized by a posterior PDF that is updated recursively as measurements are acquired. To this end, the forward encoding-detection process of GI is formulated to construct the measurement-likelihood function $p(\mathbf{z}|\mathbf{x})$, specifically as [20,39]

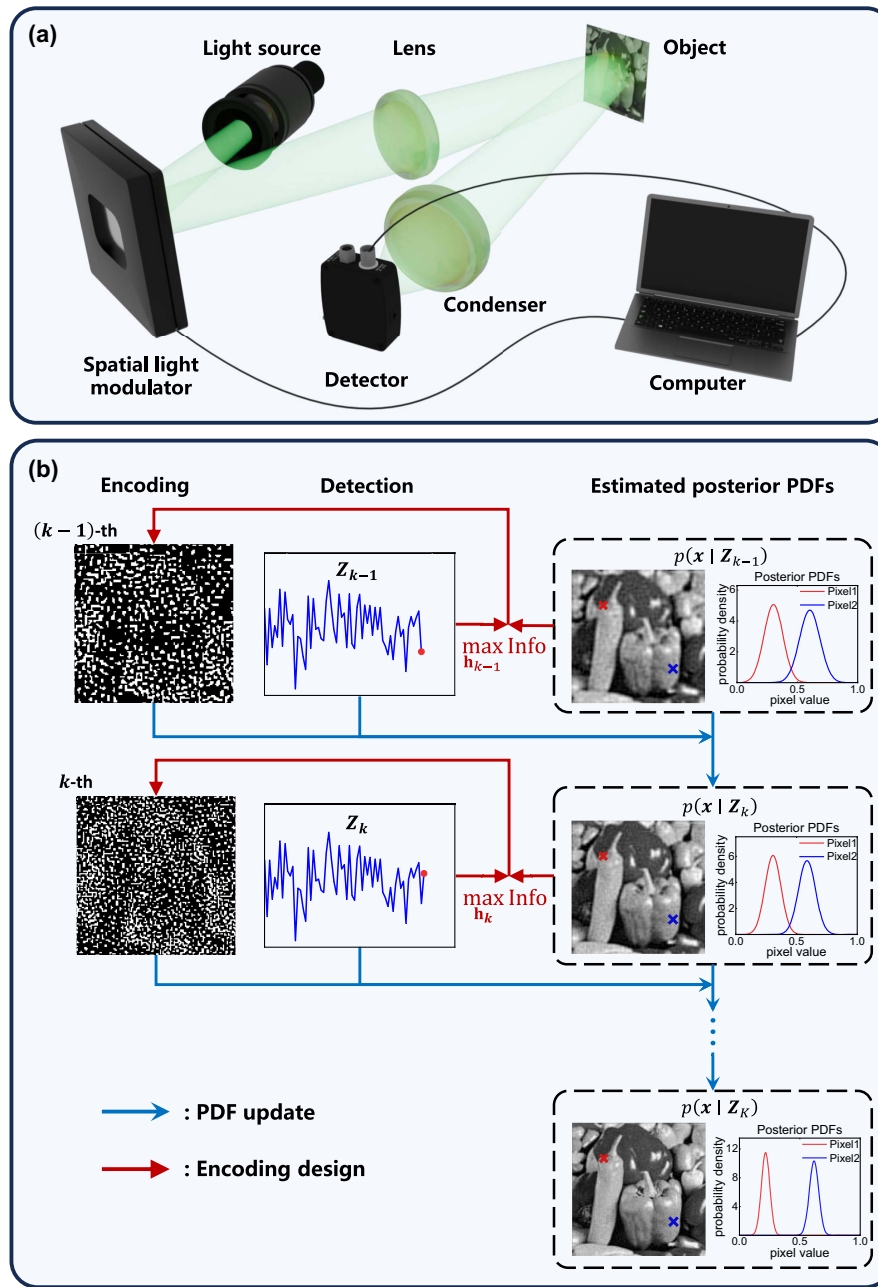


Fig. 1. (a) Schematic illustration of the ghost imaging (GI) system based on a spatial light modulator. (b) Overview of the proposed adaptive information-maximization encoding (AIME) framework for GI. The optimal encoding is designed by maximizing the information contained in the detection signal of the object to be imaged. The information is calculated based on the forward detection model and posterior PDFs.

$$\mathbf{z} = \beta \mathbf{H} \mathbf{x} + \mathbf{n}, \quad (1)$$

where $\mathbf{z}$ stands for the recorded signal from a series of bucket detection, $\mathbf{x}$ denotes the desired object's image, $\mathbf{H}$ is the measurement matrix consisting of the encoding light fields $\{\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_K\}^T$ with K being the number of acquired measurements, $\mathbf{n}$ is the detection noise, and β represents a coefficient accounting for various system-dependent factors. This formulation is also shared by a broad class of computational imaging systems.

Then, an initial Gaussian distribution $p(\mathbf{x} | \mathbf{Z}_0) = p_0(\mathbf{x}) = \mathcal{N}(\hat{\mathbf{x}}_0, \hat{\mathbf{P}}_0)$ is assumed by incorporating statistics of imaging scenes in the frequency domain [40] (see Section 4 for more details). By considering the forward sub-model with a frame of light fields $\mathbf{h}_k$ constructed as $p(\mathbf{z}_k | \mathbf{x}) = \mathcal{N}(\beta \mathbf{h}_k^T \mathbf{x}, R_k)$, the posterior PDF after the k -th measurement is obtained as

$$p(\mathbf{x} | \mathbf{Z}_k) = \mathcal{N}(\hat{\mathbf{x}}_k, \hat{\mathbf{P}}_k). \quad (2)$$

It should be noted that, in the present implementation, both the object posterior and the measurement noise are treated under Gaussian approximations. Specifically, the noise model

is adopted as $n_k \sim \mathcal{N}(0, R_k)$, where $R_k = \omega^2 \mathbf{h}_k^\top \hat{\mathbf{x}}_{k-1}$ results from the photon noise assumption [39], with the current estimate $\hat{\mathbf{x}}_{k-1}$ used to approximate the unknown true $\mathbf{x}$. This approximation enables efficient recursive updating and provides analytical posterior statistics for the subsequent information-theoretic encoding optimization. Under these settings, the posterior statistics $\hat{\mathbf{x}}_k, \hat{\mathbf{P}}_k$ are obtained via recursive expressions:

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_{k-1} + \mathbf{K}_k(\mathbf{z}_k - \beta \mathbf{h}_k^\top \hat{\mathbf{x}}_{k-1}), \quad (3a)$$

$$\hat{\mathbf{P}}_k = (\mathbf{I} - \beta \mathbf{K}_k \mathbf{h}_k^\top) \hat{\mathbf{P}}_{k-1}, \quad (3b)$$

with

$$\mathbf{K}_k = \beta \hat{\mathbf{P}}_{k-1} \mathbf{h}_k (\beta^2 \mathbf{h}_k^\top \hat{\mathbf{P}}_{k-1} \mathbf{h}_k + R_k)^{-1}.$$

These posterior statistics serve as sufficient descriptors of the statistical state and provide the basis for quantitatively assessing the information content of subsequent measurements, as will be used in the following subsection.

2. Theoretical Objective of AIME

Based on the Gaussian-approximated posterior PDF $p(\mathbf{x}|\mathbf{Z}_k)$ obtained from recursive inference, the encoding for the $(k+1)$ -th measurement can be formulated as a variable to be optimized according to an information-theoretic criterion. In the AIME framework, this optimization is guided by the expected reduction of posterior uncertainty induced by a candidate encoding, which provides a principled definition of information-optimal encoding. Here, the mutual information and the Fisher-information-related criterion are employed as two representative information-theoretic measures of this principle.

- The mutual information between the k -th detection signal and the imaging object $I(\mathbf{z}_k, \mathbf{x}|\mathbf{Z}_{k-1})$ is calculated by

$$I(\mathbf{z}_k, \mathbf{x}|\mathbf{Z}_{k-1}) = \int d\mathbf{z}_k d\mathbf{x} p(\mathbf{z}_k, \mathbf{x}|\mathbf{Z}_{k-1}) \log \frac{p(\mathbf{z}_k, \mathbf{x}|\mathbf{Z}_{k-1})}{p(\mathbf{z}_k)p(\mathbf{x}|\mathbf{Z}_{k-1})} \quad (4)$$

and obtained as

$$\begin{aligned} I(\mathbf{z}_k, \mathbf{x}|\mathbf{Z}_{k-1}) &= \frac{1}{2} \log \frac{1}{|\mathbf{I} - \beta \mathbf{K}_k \mathbf{h}_k^\top|} \\ &= \frac{1}{2} \log \left(1 + \beta^2 \frac{\mathbf{h}_k^\top \hat{\mathbf{P}}_{k-1} \mathbf{h}_k}{R_k} \right). \end{aligned} \quad (5)$$

Hence, maximizing the mutual information is equivalent to maximizing the following objective:

$$\mathcal{L}_{\text{MI}} = \frac{\mathbf{h}_k^\top \hat{\mathbf{P}}_{k-1} \mathbf{h}_k}{R_k}. \quad (6)$$

For convenience, this mutual-information-based formulation is hereafter referred to as AIME-MI.

- The Fisher information contained in the detection signal $\mathbf{z}_k$ is in fact represented by the Fisher information matrix (FIM) $\mathbf{P}_k^{-1}$ [33]. Since the calculation of matrix inverse is time-consuming, instead of FIM, here the Cramér-Rao bound (CRB) is applied to perform the information-optimal encoding design by minimizing it. Specifically, the mean CRB (mCRB), which stands for the error bound averaging on all pixels of the imaging object, is calculated by

$$\text{mCRB} = \text{Tr}(\hat{\mathbf{P}}_k) \quad (7)$$

and is obtained as

$$\text{mCRB} = \text{Tr}(\hat{\mathbf{P}}_{k-1}) - \frac{\mathbf{h}_k^\top \hat{\mathbf{P}}_{k-1}^2 \mathbf{h}_k}{\mathbf{h}_k^\top \hat{\mathbf{P}}_{k-1} \mathbf{h}_k + \frac{R_k}{\beta^2}}. \quad (8)$$

Since the covariance matrix $\hat{\mathbf{P}}_{k-1}$ has been determined by already-acquired detection, to minimize the mean CRB, only the last term in Eq. (8) can be optimized. Hence, minimizing the mCRB is equivalent to maximizing the following objective:

$$\mathcal{L}_{\text{CRB}} = \frac{\mathbf{h}_k^\top \hat{\mathbf{P}}_{k-1}^2 \mathbf{h}_k}{\mathbf{h}_k^\top \hat{\mathbf{P}}_{k-1} \mathbf{h}_k + \frac{R_k}{\beta^2}}. \quad (9)$$

For convenience, this formulation is hereafter referred to as AIME-CRB.

With Eqs. (6) and (9), it is possible to optimize the subsequent encoding $\mathbf{h}_k$ by

$$\max_{\mathbf{h}_k} \mathcal{L}, \text{ s.t. } \mathcal{C}(\mathbf{h}_k), \quad (10)$$

where $\mathcal{C}(\mathbf{h}_k)$ denotes the practical constraints on the encoding $\mathbf{h}_k$. We would like to note that, although derived from an estimation-theoretic perspective, the CRB-based criterion quantifies the same objective of posterior uncertainty reduction as the mutual-information-based formulation.

Under this formulation, the design of information-optimal encoding is unified as a constrained optimization problem driven by posterior PDF. Once a specific information criterion and the physical constraints on the encoding are specified, the structure of the optimal encoding follows directly from the solution of this optimization problem. Consequently, the AIME framework itself is not tied to a particular form of information measure or constraint.

B. Information-Optimal Encoding in an Ideal Energy-Constrained Regime

Based on the information-theoretic objective derived in the previous subsection, we first examine AIME in an idealized yet physically meaningful encoding regime. Specifically, we consider a total-energy constraint on the encoding light fields, which captures a fundamental limitation in optical imaging systems. Since the total energy of encoding $\mathbf{h}_k$ can be represented by its ℓ_1 norm, i.e., $\sum_j \mathbf{h}_{kj} = \|\mathbf{h}_k\|_1$, AIME under this constraint is denoted with the suffix “- ℓ_1 ”. In this regime, the optimal encoding admits a closed-form analytical solution, as established in the following theorem.

Theorem 1 (Theoretical solution of information-optimal encoding under a total-energy constraint). Consider the following regularized optimization problem:

$$\max_{\mathbf{h}_k} \frac{\mathbf{h}_k^\top \hat{\mathbf{P}}_{k-1} \mathbf{h}_k}{\mathbf{h}_k^\top \mathbf{x} + \epsilon}, \text{ s.t. } \|\mathbf{h}_k\|_1 = C, \quad \mathbf{h}_k \geq 0, \quad (11)$$

where $\hat{\mathbf{P}}_{k-1}$ is a positive definite covariance matrix, $\mathbf{x}$ is a non-negative vector, $C > 0$ is a constant, and ϵ is a small positive regularization term introduced to avoid a zero denominator. Then the above problem admits an analytical optimal solution. Specifically, the optimal encoding vector is given by

$$\hat{\mathbf{h}}_k = C\mathbf{e}_{\hat{i}}, \quad \hat{i} = \arg \max_i \frac{(\mathbf{P}_{k-1})_{ii}}{x_i + \epsilon/C}, \quad (12)$$

where $\mathbf{e}_i$ denotes the standard basis vector whose i -th element is 1 and all other elements are 0.

This theorem establishes that, in the ideal energy-constrained regime, information-optimal encoding necessarily reduces to an adaptive point-wise scanning strategy. This point-wise structure does not imply a predetermined or fixed scanning order, but instead, varies according to the evolving posterior statistics. Physically, the mutual information objective maximizes the information gained from the next scalar bucket measurement, which is related to how reliably the uncertainty-induced variation in the bucket signal can be distinguished from the detection noise. Under a fixed total-energy constraint, though distributing illumination over multiple pixels collects mixed information from several pixels, the modulation strength assigned to each pixel is weakened. Since the contribution of each pixel to the uncertainty-induced variation scales with the square of its modulation intensity, it is more efficient to concentrate the available energy on the currently most informative pixel, leading to a point-like optimal encoding pattern.

For the CRB-based objective shown as Eq. (9), although a closed-form analytical solution of the optimal encoding under the total-energy constraint is difficult to derive, simulation and experimental results show that the optimal encoding initially appears as a diffuse illumination pattern and gradually concentrates into a localized spot as the number of acquired measurements increases. This behavior can be understood from its objective of reducing the total posterior estimation variance after the next measurement. At early stages, when the uncertainty is broadly distributed over many spatial pixels, a spatially extended pattern can jointly probe multiple uncertain components and reduce the total estimation variance more effectively. As the posterior uncertainty becomes progressively reduced and spatially localized, the AIME- ℓ_1 -CRB pattern gradually concentrates on the remaining high-uncertainty region.

Figure 2 provides an illustrative visualization of the above structural evolution of AIME- ℓ_1 encoding patterns, where Figs. 2(a) and 2(b) correspond to two representative imaging objects. To ensure a fair comparison, the first encoding pattern is identically initialized as the first Hermite Gaussian mode [41] for all cases. The results visually corroborate the theoretical prediction that information-optimal encoding evolves toward an adaptive point-wise structure under the total-energy constraint, with the sampling order governed by the posterior statistical uncertainty rather than being fixed in advance.

We then evaluate the imaging performance of the AIME- ℓ_1 framework. In this experiment, AIME- ℓ_1 is implemented within the proposed Bayesian framework, whereas conventional imaging is obtained directly from fixed raster-scan measurements. To further separate the effect of adaptive sampling from that of Bayesian filtering, we also include random and sequential point-scanning measurement strategies with the same Bayesian filtering reconstruction. Details of the experimental setting for this comparison are presented in Section 4.C. The experimental results are shown in Fig. 3. Figures 3(a1) and 3(b1) present representative reconstructed images of two scenes under SRs of 50% and 100%,

respectively, with two detection SNR levels shown for each scene. Figs. 3(a2), 3(a3), 3(b2), and 3(b3) show quantitative evaluation of imaging quality in terms of PSNR and SSIM as functions of the SR, respectively. It can be observed in both scenes that AIME- ℓ_1 achieves a higher imaging quality than conventional direct point-scanning imaging, especially under low-SNR and undersampled conditions. For example, at 50% SR, AIME- ℓ_1 already provides more recognizable structural information than conventional direct imaging obtained at full sampling under the same SNR. Besides, results corresponding to random point scanning strategy also provide strong reconstruction performance and can approach the quality of conventional full-sampling point-scanning imaging at relatively low SRs, indicating that a major part of the improvement over conventional imaging comes from Bayesian-filtering-based utilization of undersampled point-wise measurements. In comparison, AIME- ℓ_1 provides a further but moderate improvement in the low-SR region. This can also be supported by the acquired-information curves shown in Figs. 3(a4) and 3(b4), where AIME- ℓ_1 accumulates slightly more information than the random point-scanning mode under the same SR. Therefore, the contribution of AIME- ℓ_1 should be understood at two levels: it theoretically identifies point-wise measurement as the information-optimal form under the total-energy constraint, and further optimizes the sampling order according to the evolving posterior statistics.

An alternative intuition for the emergence of point-wise optimal encoding can be drawn from CS. It is well established in CS theory that sparse signals can be efficiently acquired through incoherent measurements in a conjugate domain, for example, spatially sparse signals can be recovered from partial Fourier measurements [42]. In the present imaging scenario, natural scenes are often sparse in transform domains such as the discrete cosine transform (DCT) domain. From this perspective, concentrating measurements on a limited set of spatial locations can be viewed as an intuitive counterpart to selectively probing sparse degrees of freedom in a conjugate domain, offering a qualitative analogy to the point-wise scanning behavior predicted by the information-theoretic analysis.

C. Information-Optimal Encoding under Realistic Modulation Constraints

The preceding analysis establishes the intrinsic structure of information-optimal encoding under an idealized total-energy constraint. In practical GI systems, however, additional modulation constraints on the encoding patterns are inevitably imposed. It is therefore essential to examine whether the advantages of AIME persist under such realistic conditions. To this end, beyond the ideal total-energy constraint, we further impose a bounded-amplitude constraint on the encoding patterns,

$$0 \leq h_{k,i} \leq C, \quad \forall i, \quad (13)$$

which reflects realistic hardware limitations such as the finite modulation depth of spatial light modulators or digital micro-mirror devices. To differentiate with AIME- ℓ_1 , AIME under this constraint is denoted with the suffix “ $-\ell_\infty$ ”, since the maximal element of non-negative encoding $\mathbf{h}_k$ can be represented by its ℓ_∞ norm.

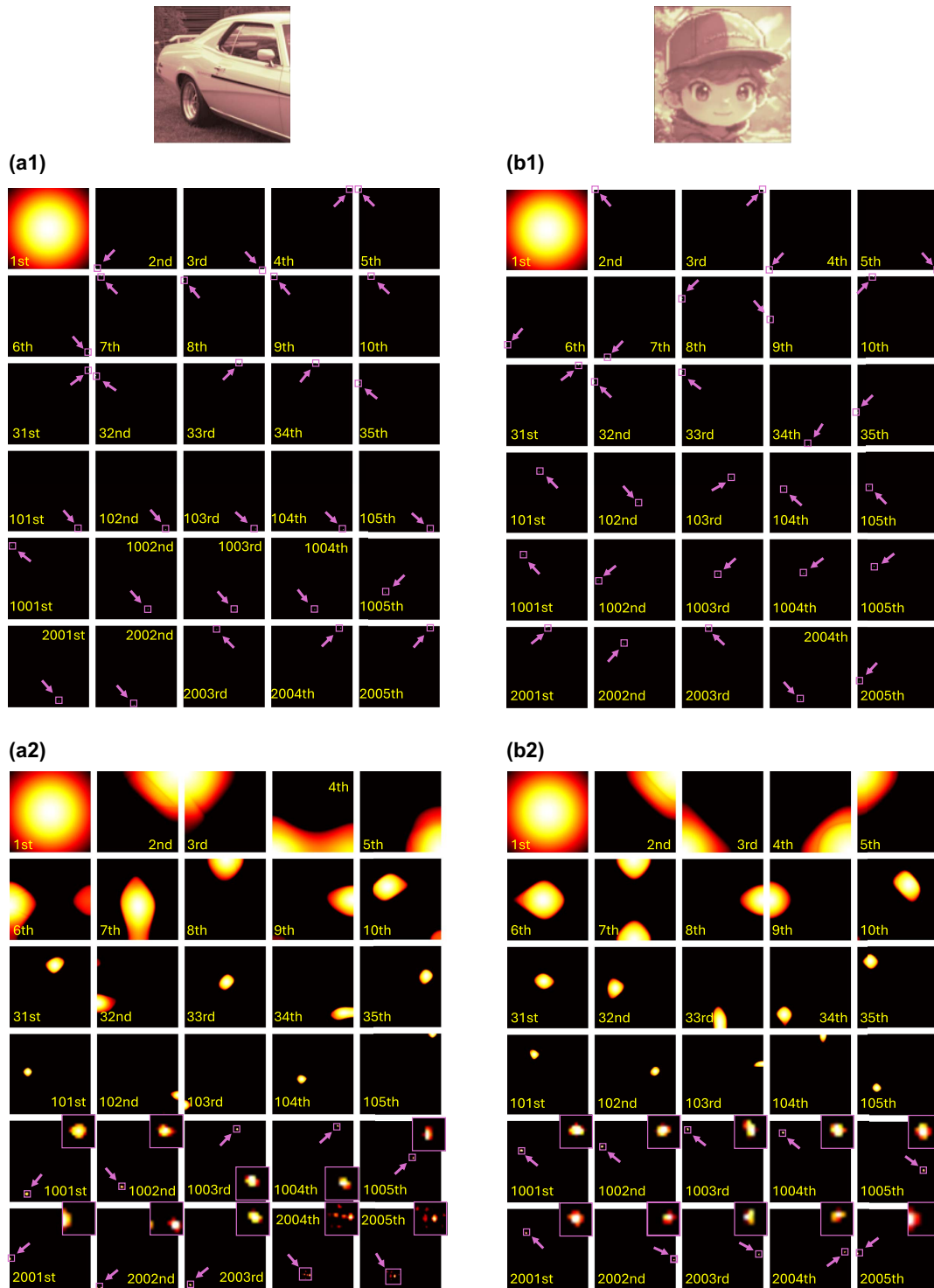


Fig. 2. Illustration of the theoretical result of AIME- ℓ_1 patterns; each pattern is normalized to its maximum value for visualization. (a1), (a2) and (b1), (b2) Respectively, AIME results of two different imaging objects. (a1), (b1) Patterns obtained by analytical solution of Eq. (12), each of which shows exactly a scanning point; and the position of the point is marked with a purple arrow and box. (a2), (b2) Patterns obtained by minimizing the mean CRB, which gradually become a small-size spot as the number of measurements increases; and the spot region in each pattern is displayed in an enlarged view. The first Hermite-Gaussian pattern is employed as the first frame of encoding pattern here for all cases.

Under such constraints, the closed-form analytical solution derived above no longer applies, and the information-optimal encoding is therefore obtained numerically by solving the resulting constrained optimization problem. Figure 4 provides

an illustrative visualization of the joint evolution of imaging results and the optimized encoding patterns under the proposed strategy. Two different imaging scenes, namely, cameraman and pepper, are considered, and the PSNR is used

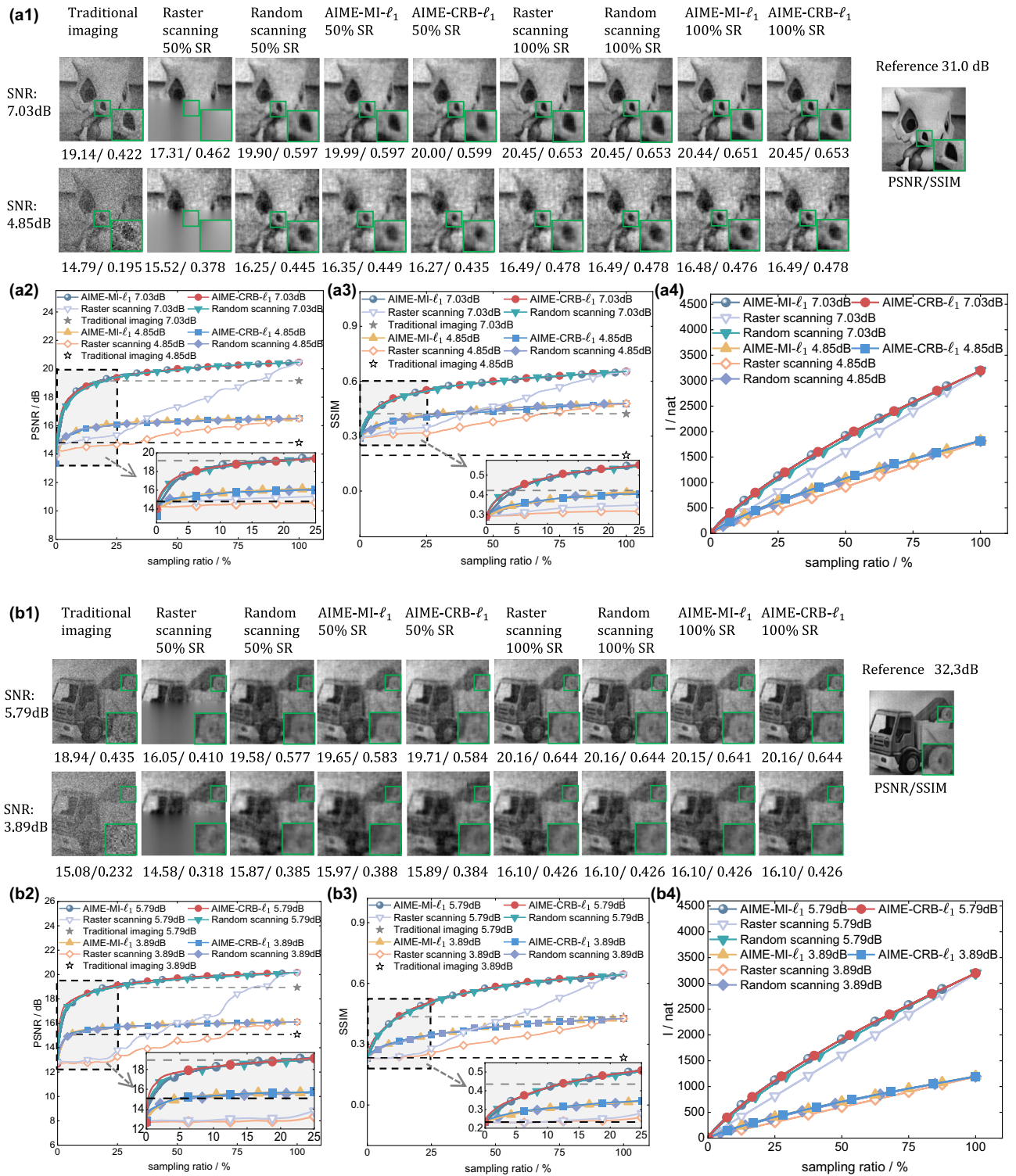


Fig. 3. Experimental comparison of imaging results obtained by GI with AIME- ℓ_1 , other point-scanning strategies, and conventional imaging. (a1), (b1) Imaging results of two representative scenes obtained under different signal-to-noise ratio (SNR) levels. Results are shown for conventional direct point-scanning imaging, random point scanning with Bayesian filtering, sequential point scanning with Bayesian filtering, and GI with AIME- ℓ_1 at representative sampling ratios (SRs). The reference image shown on the right is obtained by full-sampling point-wise scanning under a significantly higher SNR. (a2), (a3), (b2), (b3) Quantitative evaluation of imaging quality in terms of PSNR and SSIM as functions of the SR, respectively. The horizontal lines indicate the performance of conventional full-sampling imaging for comparison, and the insets enlarge the low-SR regions. (a4), (b4) Accumulated information acquired by different point-scanning strategies.

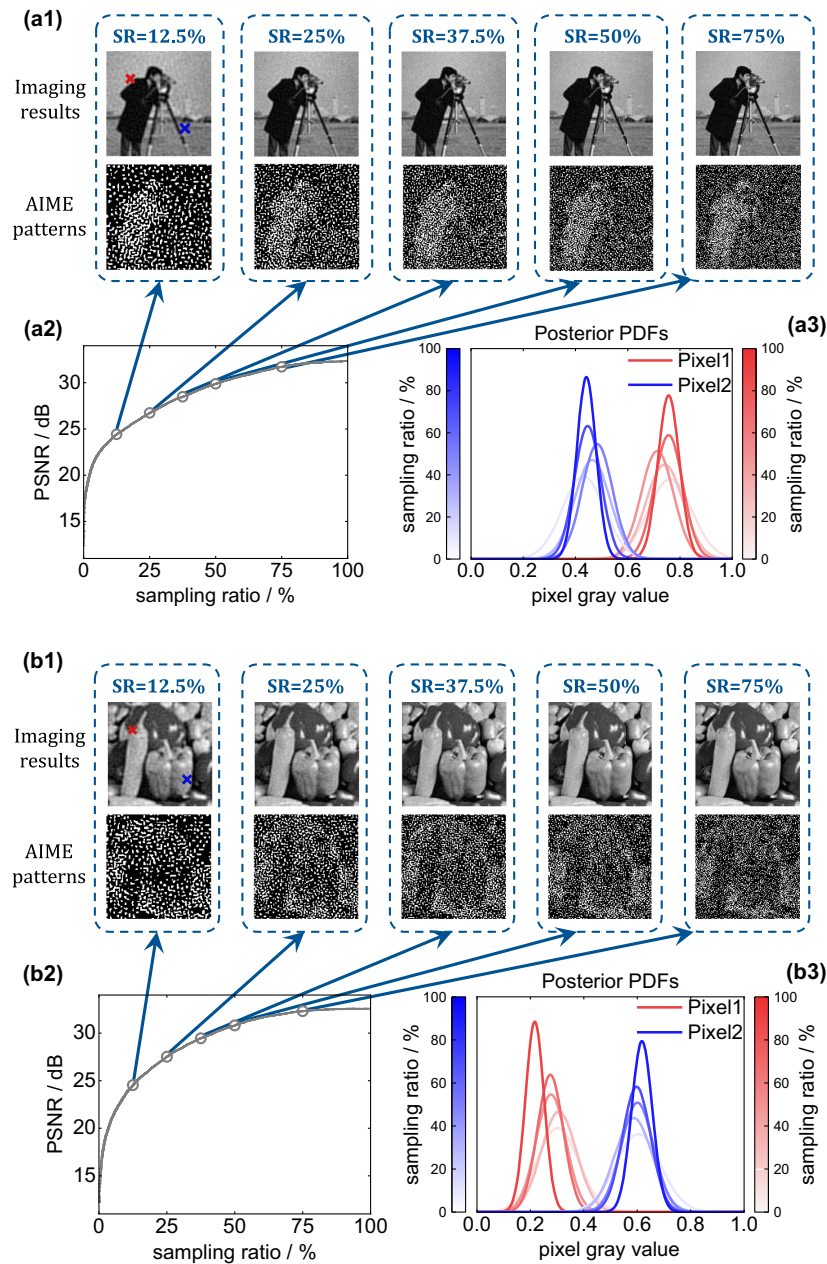


Fig. 4. Simulation results for illustrating the evolution process of the proposed strategy on two different imaging scenes. (a1), (b1) Evolution of estimated images and corresponding optimized encoding patterns with sampling ratio increasing for two imaging scenes, respectively; each displayed pattern corresponds to a single optimized encoding pattern at the indicated measurement step; (a2), (b2) evolution of PSNR indicators with the increase in sampling; (a3), (b3) evolution of estimated PDFs of two selected pixels as marked on the imaging results.

to quantify the imaging quality. Figures 4(a1) and 4(b1) show imaging results together with the corresponding optimized encoding patterns at different sampling ratios (SRs), where the selected SRs correspond to the marked points in Figs. 4(a2) and 4(b2). The optimized encoding patterns exhibit a clear coarse-to-fine evolution as the SR increases. At low SRs, the patterns consist of large-scale speckle-like structures, whereas at higher SRs, finer spatial features gradually emerge. This behavior is consistent with the gradual refinement of the posterior distribution in the Bayesian filtering process, where early measurements focus on coarse information content and

subsequent measurements concentrate on increasingly localized spatial details. The corresponding PSNR curves shown in Figs. 4(a2) and 4(b2) demonstrate a rapid improvement in reconstruction quality at the early stage of sampling, indicating the high information efficiency of the proposed strategy under limited measurements. To further illustrate the accuracy and uncertainty evolution of the reconstruction, the posterior probability density functions (PDFs) of two representative pixel locations (marked by red and blue crosses in the estimated images) are plotted in Figs. 4(a3) and 4(b3). As SR increases, the posterior PDFs become progressively sharper, reflecting the

systematic reduction of estimation uncertainty and the convergence of pixel-wise intensity estimates. Finally, by comparing the results obtained for the two different imaging scenes, it is evident that the optimized encoding patterns are also strongly scene-dependent under the bounded-amplitude constraint. This observation is fully consistent with the theoretical formulation of AIME and highlights its ability to adaptively allocate measurements according to the estimated statistical state of the imaging object.

To evaluate the practical effectiveness of AIME- ℓ_∞ under such realistic experimental constraints, we compare it with several representative existing encoding patterns in GI, including pseudo-thermal random patterns, Walsh-Hadamard patterns, and Hermite-Gaussian patterns [41]. It should be noted that many existing GI methods achieve performance improvements through advances in the reconstruction stage. Directly comparing AIME with them would mix the effects of encoding design and reconstruction algorithms. Therefore, all compared encoding strategies are evaluated under the same Bayesian filtering reconstruction framework. For theoretical reference, we also include the ideal total-energy-constrained AIME- ℓ_1 results, which serve as a theoretical performance limit without experimental modulation constraints.

Figure 5 summarizes the results of three representative scenes acquired under different SRs and SNRs. The SR and SNR for them are as follows: Fig. 5(a1), SNR around 18 dB, and 25% SR; Fig. 5(b1), SNR around 18.5 dB, and 37.5% SR; Fig. 5(c1), SNR around 20 dB, and 37.5% SR. The imaging results are shown as Figs. 5(a1), 5(b1), and 5(c1), respectively. For all tested scenes, both AIME-based strategies (i.e., AIME-MI and AIME-CRB) consistently yield superior imaging quality compared with other encoding patterns under the same modulation constraints, as evidenced by clearer structural details and significantly reduced noise artifacts. More quantitatively, the corresponding PSNR and SSIM values are reported under each image, which confirm that AIME achieves much better quantitative performance across all conditions. Specifically, at the representative sampling conditions shown in Figs. 5(a1), 5(b1), and 5(c1), the best fixed encoding baseline is Hermite-Gaussian encoding. Under the same sampling and noise conditions, the experimentally constrained AIME- ℓ_∞ results yield PSNR gains of 2.56–3.91 dB and SSIM improvements of 0.039–0.144. At the same time, a performance gap between AIME- ℓ_∞ and AIME- ℓ_1 theoretical reference can be observed, suggesting the loss of achievable information imposed by hardware constraints on the encoding space. The evolution of PSNR and SSIM as functions of the SR is further plotted in Figs. 5(a2), 5(a3), 5(b2), 5(b3), and 5(c2), 5(c3), respectively. Additional reconstructed images in the ultra-low-SR region are provided in Fig. 8, Appendix D to further visualize the reconstruction behavior corresponding to the quantitative curves. These results reveal that AIME- ℓ_∞ exhibits a pronounced advantage at early measurement stages, where substantial imaging quality improvements are already achieved at low SRs. As the number of measurements increases, AIME- ℓ_∞ continues to provide steady performance gains, whereas the imaging quality obtained with other encoding patterns rapidly saturates. This behavior demonstrates the strong compressed-

sampling capability of AIME and its effectiveness in allocating the most informative measurements. The gap between AIME- ℓ_∞ and AIME- ℓ_1 can be also reflected from imaging quality curves, which quantifies the performance loss induced by realistic modulation constraints.

To further quantify the information acquisition capability of different encoding strategies, we explicitly compute the cumulative mutual information accumulated from obtained detection signals. Specifically, the cumulative mutual information is calculated as $\sum_k \frac{1}{2} \ln(1 + \frac{\mathbf{h}_k \hat{\mathbf{P}}_{k-1} \mathbf{h}_k^T}{R_k})$ and plotted on the logarithmic scale. The corresponding results shown in Figs. 5(a4), 5(b4), and 5(c4) represent evolution of the acquired amount of information about the unknown imaging scene during the detection process. It can be found that, across different SRs and noise levels, AIME consistently enables the detection signals to acquire substantially more information about the imaging object than other comparison encoding strategies. Also, the separation between AIME- ℓ_∞ and AIME- ℓ_1 curves quantifies the information loss due to realistic modulation constraints, and mirrors the gap observed in reconstruction quality. Moreover, the ordering of the information curves agrees with the ordering of reconstruction quality, establishing a direct correspondence between information acquisition and imaging performance. This agreement confirms that the superiority of AIME arises from principled information maximization rather than other heuristic effects.

Based on the above results, it is also useful to compare the practical behaviors of the AIME-MI and AIME-CRB criteria. In theory, AIME-MI is related to expected information gain, whereas AIME-CRB targets the reduction of posterior estimation variance [43]. Therefore, consistent with previous studies [44], the two criteria should be regarded as complementary rather than universally ordered. For the GI applications considered here, the distinction is reflected in Fig. 5. Under the ℓ_1 total-energy constraint, AIME-CRB achieves better low-SR reconstruction because its distributed patterns directly reduce posterior variance. By contrast, AIME-MI reduces to point-wise information maximization, leading to weaker early-stage reconstruction but steadily improved performance that becomes more favorable at high SRs. Under the ℓ_∞ constraint, AIME-MI and AIME-CRB show similar PSNR and SSIM trends, indicating that the two criteria lead to comparable continuous-valued adaptive patterns under this constraint. The two criteria also differ in computational cost and numerical stability, especially under the ℓ_1 total-energy constraint: AIME-MI admits an analytical point-selection solution without iterative optimization, making it simpler and less sensitive to optimization settings, whereas AIME-CRB still requires numerical optimization. The detailed optimization settings and runtime measurements are provided in Section 4.A. To conclude, AIME-CRB is recommended when early-stage reconstruction quality and posterior-variance reduction are prioritized. AIME-MI is recommended when computational simplicity and information-gain acquisition are prioritized.

D. Scalability to Larger Image Sizes

The AIME framework is not intrinsically restricted to a specific image size, since both the Bayesian update and information-based

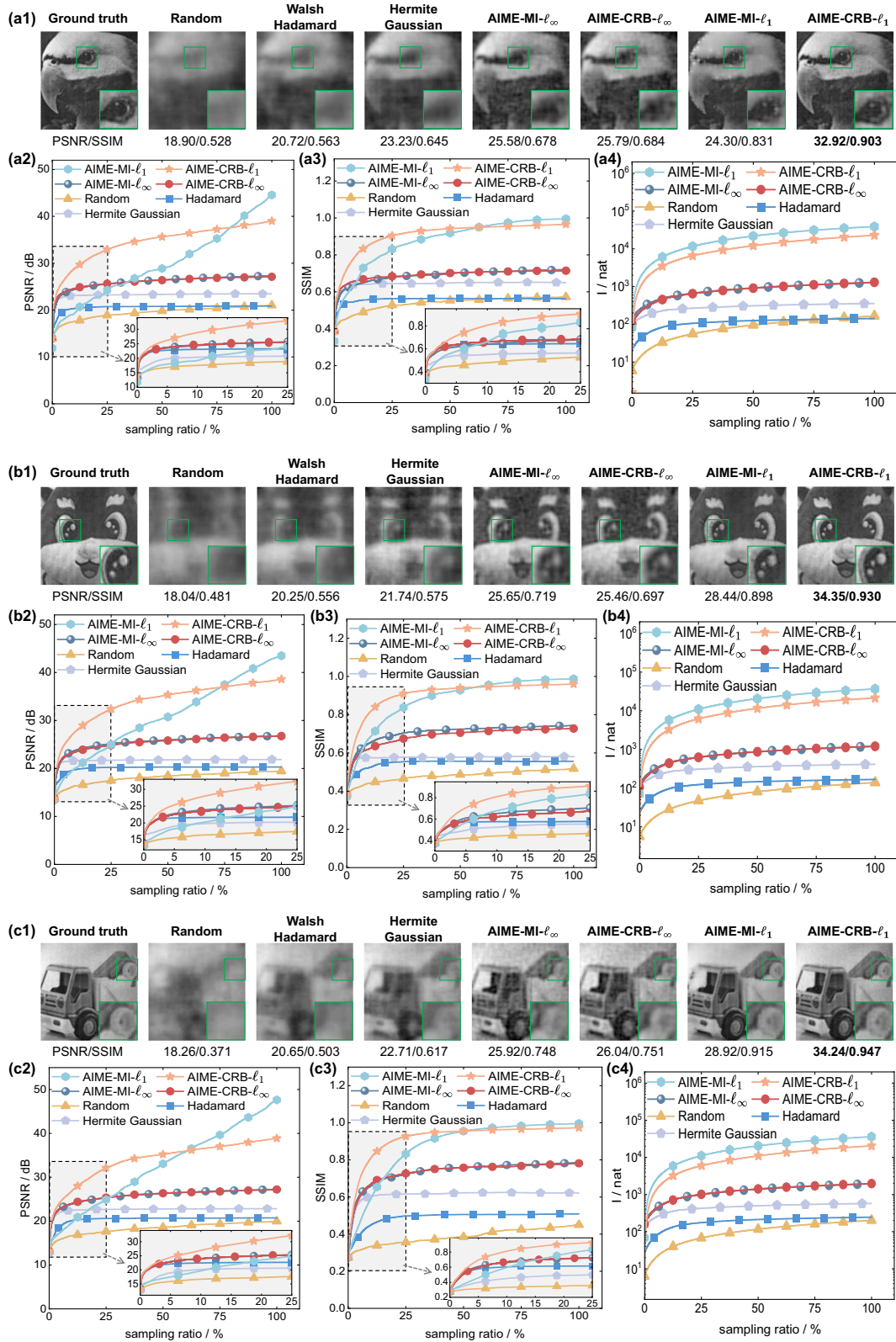


Fig. 5. Experimental imaging results for comparison between AIME- ℓ_∞ and some typical encoding patterns. Results of AIME- ℓ_1 are also shown here to serve as a fundamental-limit reference. (a1), (b1), (c1) Imaging results of three different representative scenes at selected sampling ratios (SRs) for visual comparison, where the PSNR/SSIM values below the images correspond only to the displayed SR. See Appendix D for additional imaging results under other SRs. Specifically, the SR and SNR for them are (a1) SNR around 18 dB, and 25% SR; (b1) SNR around 18.5 dB, and 37.5% SR; (c1) SNR around 20 dB, and 37.5% SR. To provide quantitative comparisons beyond these representative examples, PSNR and SSIM curves over the full SR range for each imaging scene are shown in (a2), (a3), (b2), (b3), and (c2), (c3), respectively. The low-SR region is further enlarged in the insets, highlighting the performance comparison at SRs of 25% and below. To further quantify and confirm the information acquisition enhancement, the acquired information of different encoding strategies as a function of SR is calculated, and shown in (a4), (b4), and (c4).

encoding design can be formulated for a vectorized scene with an arbitrary number of pixels. To examine its scalability, we performed additional simulations on 256×256 images under two SNR levels of 30 dB and 20 dB, as shown in Fig. 6. Figures 6(a1) and 6(b1) show that, compared with fixed encoding patterns including random, Walsh-Hadamard, and Hermite-Gaussian patterns, the AIME-based methods preserve clearer

structural details, especially at low and moderate SRs. This advantage remains observable at the lower SNR, although all methods exhibit performance degradation. The corresponding PSNR and SSIM curves in Figs. 6(a2), 6(a3), 6(b2), and 6(b3) further confirm that AIME achieves higher reconstruction quality over a broad range of SRs, with a clear advantage in the low-SR regions enlarged in the insets. These results indicate that the performance

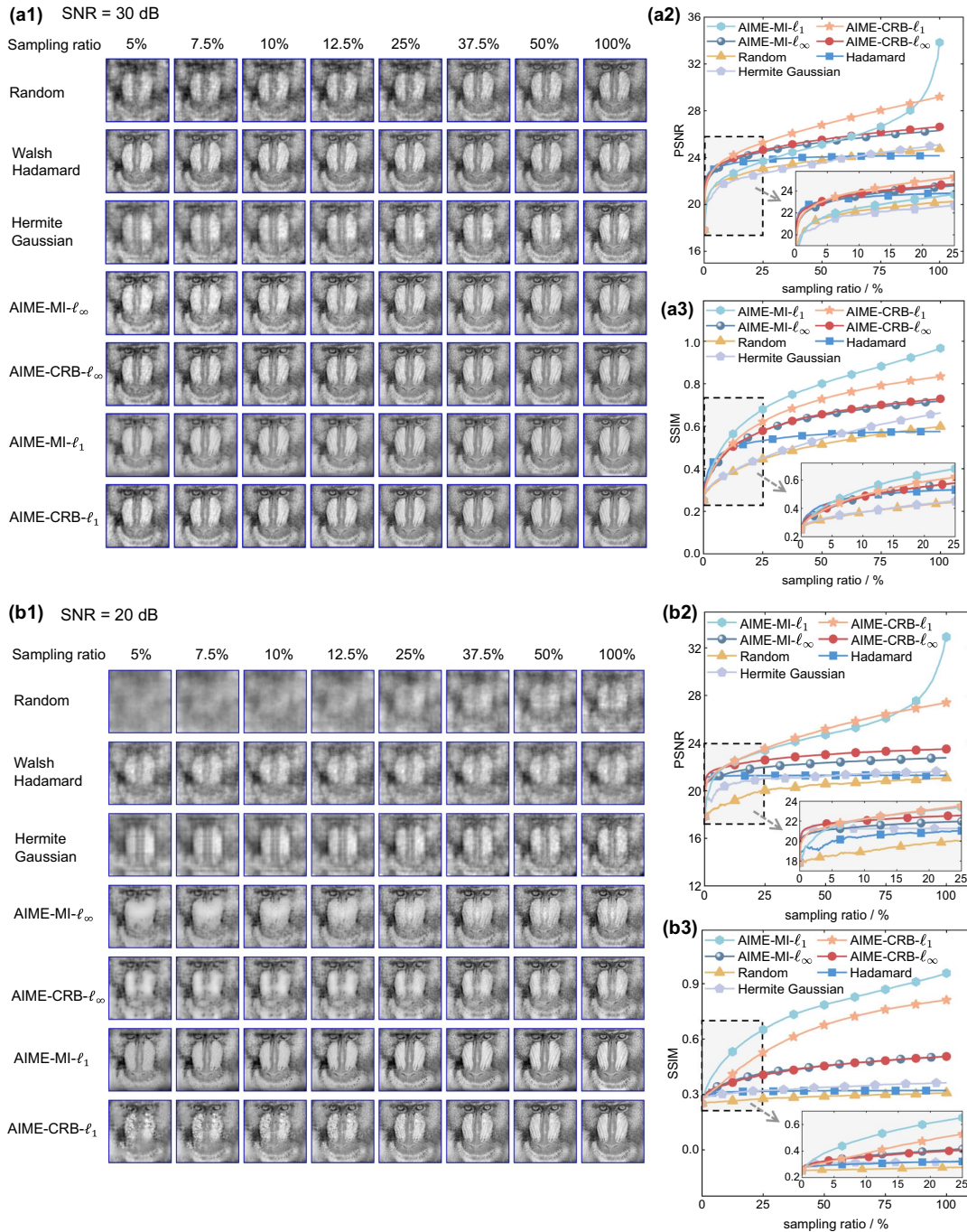


Fig. 6. Scalability evaluation of AIME for a larger image size. The image size is increased to 256×256 , and two detection SNR levels of 30 dB and 20 dB are considered. (a1), (b1) Representative reconstruction results under SNRs of 30 dB and 20 dB, respectively. In each image grid, the columns correspond to SRs from 5% to 100%, and the rows correspond to different encoding strategies. (a2), (b2) Corresponding PSNR curves as functions of the SR; (a3), (b3) corresponding SSIM curves. The low-SR regions are enlarged in the insets. The compared methods include random patterns, Walsh-Hadamard patterns, Hermite-Gaussian patterns, AIME-MI- ℓ_∞ , AIME-CRB- ℓ_∞ , AIME-MI- ℓ_1 , and AIME-CRB- ℓ_1 .

advantage of AIME is not limited to the original image scale and can extend to larger-scale images.

It is worth noting that the direct full-covariance implementation used in this work may incur rapidly increasing memory and computational costs as the image size increases, and thus may not be suitable for very large images. Nevertheless, this is an implementation limitation rather than a limitation of the AIME principle. For larger-scale applications, scalable strategies such as information-filter-based formulations [45], diagonal or block-diagonal covariance approximations [46], and low-rank covariance approximation [47] may be explored to reduce the cost of posterior updating and encoding optimization while preserving the adaptive information-driven measurement design. In addition, Kalman-filter-inspired neural network implementations [48] may provide a potential direction for future acceleration.

3. DISCUSSION

In this work, an adaptive information-maximization encoding strategy is developed for GI. The proposed framework is intended to provide a general Bayesian formulation for information-optimal encoding in computational imaging systems with the encoding-decoding mode, provided that the forward imaging model can be appropriately specified. In the present study, AIME is demonstrated using a simplified linear-Gaussian forward model in a spatial-domain GI system with sequential bucket detection, which enables analytical tractability and reproducible comparison between different encoding strategies. In practical complex scenarios, the forward model may involve more complicated physical processes, and model mismatch may arise from calibration errors, unknown interference, high-dynamic-range responses, or inaccurate noise statistics. Accurate noise modeling and system calibration are therefore essential for reliable model-based imaging, as also emphasized in recent photon-limited imaging studies [49]. Since such mismatch can affect both Bayesian posterior updating and subsequent adaptive encoding design in AIME, we further test noise-statistics mismatch and forward-model calibration mismatch, as detailed in Appendix E. The results show that AIME retains advantages over representative fixed encoding baselines under moderate mismatch, while its performance gain is indeed reduced under severe mismatch. These indicate that the current AIME implementation has a certain degree of robustness to moderate parameter mismatch, while also confirming its model-based nature and its dependence on accurate calibration. Therefore, extending AIME to real-world applications requires incorporating the corresponding physical forward model [39], noise/interference model [49,50], and application-dependent prior information [51], as well as performing systematic validation under realistic imaging conditions.

Besides, since Bayesian filtering theory is not restricted to linear systems, the proposed strategy can, in principle, be generalized to computational imaging problems governed by non-linear forward models, such as the coherent detection GI [7]. Beyond the form of the forward model, the framework is also not restricted to a specific sensing configuration. Although AIME is demonstrated with sequential bucket detection, the

underlying Bayesian filtering and information optimization principles can be naturally extended to batch-measurement settings. In such cases, multiple measurements are acquired simultaneously, and the framework can be generalized to adaptive or semi-adaptive encoding design in a higher-dimensional measurement space. This suggests its applicability to a broader class of computational imaging systems with array detection, such as the GI camera [9] and lensless imaging [52].

In addition to the forward model, another essential component of the proposed framework is the setting of the initial image PDF. In the presented study, a general Gaussian prior is adopted by imposing constraints on the image frequency domain, which allows an analytical PDF to be defined for a completely unknown imaging scene. For imaging scenes with multimodal or non-Gaussian statistics, more expressive prior/posterior models would be needed to capture higher-order statistical properties of natural images [53]. Furthermore, with the development of deep neural models, image priors can be effectively learned from data using deep generative models. In scenarios where a sufficiently large and representative training dataset is available, the proposed strategy can be naturally combined with image priors provided by trained neural models [54,55]. Such learned priors are expected to offer a more expressive statistical representation than the general analytical prior employed in this work, potentially leading to further performance improvements. Accordingly, more advanced analytical or numerical Bayesian inference schemes, such as particle filtering [56], Gaussian-mixture filtering [57], and variational inference [58] would be required.

Altogether, this work highlights the potential of information-theoretic encoding design as a principled approach to GI and, more broadly, to computational imaging systems. By explicitly modeling the information carried by detection signals under realistic physical constraints imposed by the imaging system, the proposed framework provides a systematic methodology to characterize and approach fundamental imaging limits. The emergence of point-like encoding as information-optimal under a total-energy constraint offers an information-theoretic insight suggesting that point-to-point sampling strategies observed in biological vision [59] may reflect a principle of efficient information acquisition. Beyond GI, the proposed framework is expected to inspire further information-theoretic studies on the evaluation and optimization of imaging and sensing systems, with potential implications for a broad range of applications, including autonomous sensing, machine vision, and data-efficient perception tasks. Importantly, in numerous such systems, flexible light-field modulation platforms such as metasurfaces and programmable photonic devices are increasingly employed as core encoding components, providing a structured design space for encoding optimization. Incorporating this into the proposed framework is supposed to offer a principled pathway that integrates emerging hardware platforms with the information-optimal encoding design.

4. METHODS

A. Details for Encoding Optimization

The encoding optimization problem is formulated as a constrained optimization task and solved using projected gradient

descent (PGD), which alternates between gradient descent updates and projection onto the feasible constraint set. During the gradient descent step, we employ the SGD optimizer with learning rates of 1×10^9 and 1×10^5 for the MI-based and CRB-based objectives, respectively. All remaining optimizer hyperparameters follow the default PyTorch configuration. To balance computational efficiency and estimation accuracy, each encoding optimization is performed using 50 iterations. All numerical calculations are implemented in PyTorch 2.5.1 with CUDA 12.1 and executed on an NVIDIA GeForce RTX 3090 GPU. For the ℓ_∞ -constrained continuous-pattern design, both AIME-MI and AIME-CRB used 50 PGD iterations and required approximately 0.13 s per optimized pattern. Under the ℓ_1 constraint, AIME-MI admits a theoretical point-selection solution without iterative PGD optimization, whereas AIME-CRB still used 50 PGD iterations and required approximately 0.14 s per optimized pattern.

B. Implementation of Pattern Update and Detection in the Experiment

AIME requires the next encoding pattern to be recalculated after each measurement based on the data already acquired, which differs from the scheme using a preloaded fixed encoding sequence. Hence, in the AIME experiment, the DMD was operated in a real-time playback mode which can display the pattern transmitted from the computer. After each measurement, the detected signal was transferred to the computer for posterior updating and next-pattern optimization. The newly calculated pattern was then sent to the DMD for the next measurement. Also, a trigger signal was sent by the computer to the detector for signal acquisition, ensuring synchronization between the displayed pattern and the recorded signal.

C. Experimental Setting in Comparison between GI with AIME- ℓ_1 and Conventional Imaging

For experimental comparison between GI with AIME- ℓ_1 and “fixed point-to-point” conventional imaging, both strategies are implemented using the same DMD-based GI system to ensure identical optical and detection conditions. For AIME- ℓ_1 , the optimized encoding patterns are physically realized with the DMD. To improve illumination stability, each 4×4 block of DMD micro-mirrors is grouped to form a single effective illumination pixel. The optimized patterns are mapped onto these grouped pixels and sequentially projected onto the scene, and the corresponding bucket detection signals are recorded. For conventional imaging, the same DMD-based GI system is operated in a raster-scanning mode. Specifically, a sequence of point-like illumination patterns is loaded onto the DMD, where each grouped 4×4 block is illuminated one at a time following a fixed scanning order. The detected bucket signal associated with each illumination position is directly assigned as the grayscale value of the corresponding image pixel, thereby forming the image.

In the experiment under low-SNR conditions, additional background noise is introduced using scattering light generated by a rotating ground glass. This produces a dynamic background whose temporal fluctuation, after bucket detection and time averaging, can be well approximated as an additive noise component. The background-noise variance, the

corresponding experimental SNR, and the validation of the Gaussian approximation are described in Section 4.D.

D. Measurement of Parameters β and R_k in the Experiment

The experiment is performed with a GI system with DMD modulation. To determine the forward model in Eq. (1), the scaling factor β and the noise statistics $\{R_k\}$ are experimentally calibrated prior to imaging.

1. Calibration of β

A whiteboard with uniform reflectance is used as the calibration object. A set of M Hadamard encoding patterns $\mathbf{h}^{(1)}, \dots, \mathbf{h}^{(M)}$ is projected sequentially, and for each pattern L repeated measurements are recorded, yielding detection vectors $\mathbf{z}^{(m)} \in \mathcal{R}^L$. Then, β is estimated by

$$\beta = \left\langle \frac{\bar{z}^{(m)}}{\sum_i \mathbf{h}_i^{(m)}} \right\rangle_m, \quad (14)$$

where $\bar{z}^{(m)}$ denotes the average over elements of $\mathbf{z}^{(m)}$ and $\langle \cdot \rangle_m$ represents averaging over all calibration encoding patterns.

2. Calibration of Noise Statistics

As modeled in Section 2.A.1, for the experiments without the additional dynamic background, the measurement noise variance is parameterized as

$$R_k = \omega^2 \mathbf{h}_k^\top \mathbf{x},$$

where ω^2 characterizes the proportionality between signal intensity and noise variance and therefore controls the effective detection SNR. With those measurements on the whiteboard by using Hadamard patterns, estimation on ω^2 in practical conditions is as follows. For the m -th encoding pattern, with L repeated measurements, the corresponding noise coefficient is estimated as

$$\omega^{2(m)} = \frac{\sigma^{2(m)}}{\bar{z}^{(m)}}, \quad (15)$$

where $\sigma^{2(m)} = \frac{\sum_{l=1}^L (z_l^{(m)} - \bar{z}^{(m)})^2}{L}$ denotes the sample variance of the detection signal. The calibrated noise parameter is then taken as the ensemble average:

$$\omega_{\text{eff}}^2 = \langle \omega^{2(m)} \rangle_m. \quad (16)$$

By adjusting the illumination intensity and repeating the calibration procedure, noise parameters corresponding to different operating SNR regimes are obtained. In each regime, the calibrated ω_{eff}^2 is used consistently in both mutual-information-based and CRB-based encoding optimization.

For the low-SNR experiments described in Section 4.C, the dominant noise source is the artificially introduced dynamic background generated by the rotating ground glass. Since the fluctuation caused by this background is much stronger than the fluctuation of the target-reflected signal, the background-noise statistics are measured separately using a completely dark DMD pattern while keeping the rotating-ground-glass background enabled. To quantify the noise level, L measurements are acquired using a completely dark pattern. Let $z_{\text{back}}^{(l)}$ denote the l -th recorded signal, and $\bar{z}_{\text{back}}$ its sample

mean over L measurements; the background noise variance σ_{back}^2 is estimated as

$$\sigma_{\text{back}}^2 = \frac{\sum_{l=1}^L (z_{\text{back}}^{(l)} - \bar{z}_{\text{back}})^2}{L-1}. \quad (17)$$

With experimental detection signal $\{z_k\}$ corresponding to both AIME and raster-scanning, the experimental SNR is estimated by

$$\text{SNR} = 10 \log_{10} \left(\frac{\bar{z} - \bar{z}_{\text{back}}}{\sigma_{\text{back}}} \right), \quad (18)$$

where $\bar{z}$ is the average intensity of detection signals.

To validate the adopted Gaussian noise approximation, we further analyzed the statistics of the repeated measurements used in the above calibration procedures. For the higher-SNR calibration regime, the repeated measurement signals acquired for each Hadamard pattern were first centered by subtracting their mean and then normalized by their corresponding standard deviation. For the low-SNR regime, the background fluctuations measured with the dark DMD pattern were

processed in the same way. Empirical histograms and quantile-quantile (Q-Q) plots of the normalized samples are shown in Fig. 7, where Figs. 7(a1), 7(a2) and 7(b1), 7(b2) correspond to the higher-SNR calibration and the low-SNR background-noise condition, respectively. The measured distributions agree reasonably well with the standard normal distribution, supporting the use of an effective Gaussian noise approximation for the calibrated noise statistics.

E. Setting the Initial Image PDF

In the proposed strategy, an initial Gaussian PDF $p(\mathbf{x}) = \mathcal{N}(\mathbf{x}_0, \mathbf{P}_0)$ of the object's image is set. To specify the parameters $\mathbf{x}_0$ and $\mathbf{P}_0$, we adopt a commonly used statistical model for natural images, in which the frequency-domain coefficients $\{a(\mathbf{f})\}$ of images [40] follow a Laplacian distribution:

$$p(a(\mathbf{f})) = \frac{1}{2\lambda(\mathbf{f})} \exp \left(-\frac{|a(\mathbf{f}) - \mu(\mathbf{f})|}{\lambda(\mathbf{f})} \right), \quad (19)$$

where $\mu(\mathbf{f})$ for those non-zero-frequency components is set as 0, $\mu(\mathbf{f}_0) = \mu_0$ representing the zero-frequency amplitude is

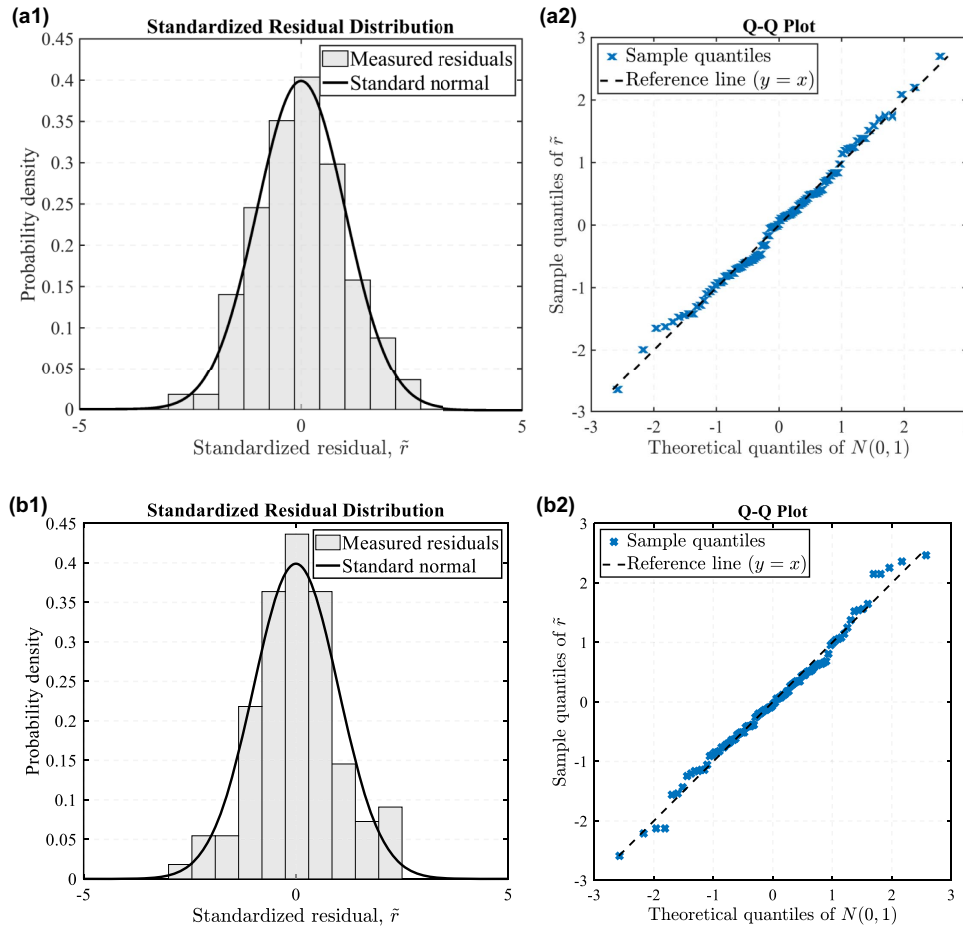


Fig. 7. Experimental validation of the Gaussian noise approximation in the calibration measurements. The histograms show the distributions of standardized repeated-measurement fluctuations, together with the standard normal distribution for comparison. The corresponding quantile-quantile (Q-Q) plots compare the sample quantiles with the theoretical quantiles of a standard normal distribution. (a1), (a2) Results under the SNR = 21.8 dB calibration condition, where the residuals are obtained from repeated measurements using a fixed whiteboard target and Hadamard encoding patterns. (b1), (b2) Results under the low-SNR condition, where the fluctuations are obtained from repeated measurements using a completely dark DMD pattern while keeping the rotating-ground-glass background enabled.

estimated with a uniform illumination, and $\lambda(\mathbf{f}) \propto (|\mathbf{f}|)^{-\gamma}$ with $\gamma \in [1, 2]$. It should be noted that, owing to the high dimensionality of image signals and the central limit theorem, although the frequency-domain coefficients are modeled as independent Laplacian variables, the image-domain pixels, being weighted superpositions of a large number of such coefficients, can be well approximated to be subject to a Gaussian distribution. Based on this, the parameters of the initial Gaussian PDF are set as

$$\mathbf{x}_0 = \mu_0 \mathbf{1}, \quad (20a)$$

$$\mathbf{P}_0 = 2\Psi^\top \Sigma^2 \Psi, \quad (20b)$$

where $\mathbf{1}$ is an all-1 vector, Σ is a diagonal matrix whose diagonal elements consist of $\{\lambda(\mathbf{f}_0), \lambda(\mathbf{f}_1), \dots\}$, and Ψ denotes the DCT matrix.

F. Source Information of the Test Images

The test images used in the reconstruction experiments include standard image-processing test images, royalty-free images, an AI-generated synthetic image, and an author-acquired photograph. The images used in Figs. 2(a), 3, and 5(a1), 5(c1) are obtained from the website <https://pixabay.com/zh/> under the Pixabay Content License, which allows free use of the content. The image used in Fig. 2(b) was generated by the authors using Doubao AI as a synthetic test image, without using any third-party copyrighted image as input. The image used in Fig. 5(b1) was acquired by the authors by photographing a real doll. The standard cameraman, peppers, and baboon images were obtained from commonly used research image databases and were used only for tests.

APPENDIX A: THEORETICAL DERIVATION OF THE OBJECTIVE FUNCTION FOR AIME-MI AND AIME-CRB

In the proposed strategy, the mutual information $I(\mathbf{z}_k, \mathbf{x}|\mathbf{Z}_{k-1})$ is specifically calculated as follows:

$$\begin{aligned} I(\mathbf{z}_k, \mathbf{x}|\mathbf{Z}_{k-1}) &= H(\mathbf{x}|\mathbf{Z}_{k-1}) - H(\mathbf{x}|\mathbf{Z}_k) \\ &= \left[\frac{N}{2} (1 + \log 2\pi) + \frac{1}{2} \log |\hat{\mathbf{P}}_{k-1}| \right] \\ &\quad - \left[\frac{N}{2} (1 + \log 2\pi) + \frac{1}{2} \log |\hat{\mathbf{P}}_k| \right] \\ &= \frac{1}{2} \log \frac{|\hat{\mathbf{P}}_{k-1}|}{|\hat{\mathbf{P}}_k|} = \frac{1}{2} \log \frac{|\hat{\mathbf{P}}_{k-1}|}{|(\mathbf{I} - \beta \mathbf{K}_k \mathbf{h}_k^\top) \hat{\mathbf{P}}_{k-1}|} \\ &= \frac{1}{2} \log \frac{1}{|\mathbf{I} - \beta \mathbf{K}_k \mathbf{h}_k^\top|}. \end{aligned} \quad (A1)$$

$H(\cdot)$ is the information entropy, which can be expressed as $\frac{1}{2} \log (2\pi e)^N \det \Sigma$ for an N -variable Gaussian distribution, Σ is the covariance matrix of the corresponding variables, and $|\mathbf{P}|$ denotes the determinant of matrix $\mathbf{P}$. By using the property of matrix determinant $|\mathbf{I} + \mathbf{u}\mathbf{v}^\top| = 1 + \mathbf{u}^\top \mathbf{v}$, the above expression can be further simplified:

$$\begin{aligned} I(\mathbf{z}_k, \mathbf{x}|\mathbf{Z}_{k-1}) &= \frac{1}{2} \log \frac{1}{1 - \beta \mathbf{K}_k^\top \mathbf{h}_k} \\ &= \frac{1}{2} \log \frac{1}{1 - (\beta^2 \mathbf{h}_k \hat{\mathbf{P}}_{k-1} \mathbf{h}_k^\top + R_k)^{-1} \beta^2 \mathbf{h}_k \hat{\mathbf{P}}_{k-1} \mathbf{h}_k} \\ &= \frac{1}{2} \log \frac{1}{1 - \frac{\beta^2 \mathbf{h}_k \hat{\mathbf{P}}_{k-1} \mathbf{h}_k^\top}{\beta^2 \mathbf{h}_k \hat{\mathbf{P}}_{k-1} \mathbf{h}_k^\top + R_k}} \\ &= \frac{1}{2} \log \frac{\beta^2 \mathbf{h}_k \hat{\mathbf{P}}_{k-1} \mathbf{h}_k^\top + R_k}{R_k}. \end{aligned} \quad (A2)$$

For the mean CRB after the k -th measurement, it can be derived and calculated as follows. Considering the Gaussian detection model $p(\mathbf{z}_k|\mathbf{x}) = \mathcal{N}(\mathbf{h}_k^\top \mathbf{x}, R_k)$, the Fisher information matrix carried by the k -th measurement for estimating the image $\mathbf{x}$ is defined as

$$\mathbf{F}_k = \mathbb{E}_{\mathbf{z}_k} \left\{ \left[\frac{\partial}{\partial \mathbf{x}} \log p(\mathbf{z}_k|\mathbf{x}, \mathbf{h}_k) \right] \cdot \left[\frac{\partial}{\partial \mathbf{x}} \log p(\mathbf{z}_k|\mathbf{x}, \mathbf{h}_k) \right]^\top \right\}, \quad (A3)$$

and can be derived as

$$\mathbf{F}_k = \beta^2 \frac{\mathbf{h}_k \mathbf{h}_k^\top}{R_k}. \quad (A4)$$

Assume that the Fisher information matrix accumulated before the k -th measurement is $\mathbf{J}_{k-1}$. Since Fisher information is additive for independent measurements, the Fisher information matrix after acquiring the k -th measurement can be recursively calculated by

$$\mathbf{J}_k = \mathbf{J}_{k-1} + \mathbf{F}_k = \mathbf{J}_{k-1} + \beta^2 \frac{\mathbf{h}_k \mathbf{h}_k^\top}{R_k}. \quad (A5)$$

According to the Bayesian filtering theory [60], the initial Fisher information $\mathbf{J}_0$ should include the information provided by the initial prior distribution, and is thus defined as

$$\mathbf{J}_0 = \mathbb{E} \left\{ \left[\frac{\partial}{\partial \mathbf{x}} \log p_0(\mathbf{x}) \right] \cdot \left[\frac{\partial}{\partial \mathbf{x}} \log p_0(\mathbf{x}) \right]^\top \right\}. \quad (A6)$$

Since the initial prior is set as a Gaussian $p_0(\mathbf{x}) = \mathcal{N}(\hat{\mathbf{x}}_0, \hat{\mathbf{P}}_0)$, one can obtain $\mathbf{J}_0 = \hat{\mathbf{P}}_0^{-1}$. By using the inductive method, assuming that $\mathbf{J}_{k-1}^{-1} = \hat{\mathbf{P}}_{k-1}$, one can obtain that

$$\begin{aligned} \mathbf{J}_k^{-1} &= \mathbf{J}_{k-1}^{-1} - \frac{\beta^2 \mathbf{J}_{k-1}^{-1} \mathbf{h}_k \mathbf{h}_k^\top R_k^{-1} \mathbf{J}_{k-1}^{-1}}{1 + \beta^2 \mathbf{h}_k^\top R_k^{-1} \mathbf{J}_{k-1}^{-1} \mathbf{h}_k} \\ &= \hat{\mathbf{P}}_{k-1} - \beta^2 \frac{\hat{\mathbf{P}}_{k-1} \mathbf{h}_k \mathbf{h}_k^\top \hat{\mathbf{P}}_{k-1}}{R_k + \beta^2 \mathbf{h}_k^\top \hat{\mathbf{P}}_{k-1} \mathbf{h}_k} = \hat{\mathbf{P}}_k. \end{aligned} \quad (A7)$$

Thus, we can conclude that the CRB matrix for estimating $\mathbf{x}$ with measurements $\{z_1, \dots, z_k\}$ is

$$\mathbf{J}_k^{-1} = \hat{\mathbf{P}}_k. \quad (A8)$$

Hence, the mean CRB is calculated as follows:

$$\begin{aligned}
\text{mCRB} &= \text{Tr}(\hat{\mathbf{P}}_k) = \text{Tr}((\mathbf{I} - \beta \mathbf{K}_k \mathbf{h}_k^\top \hat{\mathbf{P}}_{k-1}) \\
&= \text{Tr}(\hat{\mathbf{P}}_{k-1}) - \text{Tr}(\beta \mathbf{K}_k \mathbf{h}_k^\top \hat{\mathbf{P}}_{k-1}) \\
&= \text{Tr}(\hat{\mathbf{P}}_{k-1}) \\
&\quad - \text{Tr}(\beta^2 \hat{\mathbf{P}}_{k-1} \mathbf{h}_k (\beta^2 \mathbf{h}_k^\top \hat{\mathbf{P}}_{k-1} \mathbf{h}_k + R_k)^{-1} \mathbf{h}_k^\top \hat{\mathbf{P}}_{k-1}),
\end{aligned} \tag{A9}$$

where $\text{Tr}(\mathbf{P})$ denotes the trace of matrix $\mathbf{P}$. By using the property of matrix trace, the above expression can be rewritten as

$$\begin{aligned}
\text{mCRB} &= \text{Tr}(\hat{\mathbf{P}}_{k-1}) - \beta^2 \text{Tr}(\mathbf{h}_k^\top \hat{\mathbf{P}}_{k-1}^2 \mathbf{h}_k (\beta^2 \mathbf{h}_k^\top \hat{\mathbf{P}}_{k-1} \mathbf{h}_k + R_k)^{-1}) \\
&= \text{Tr}(\hat{\mathbf{P}}_{k-1}) - \text{Tr}\left(\frac{\mathbf{h}_k^\top \hat{\mathbf{P}}_{k-1}^2 \mathbf{h}_k}{\mathbf{h}_k^\top \hat{\mathbf{P}}_{k-1} \mathbf{h}_k + \frac{R_k}{\beta^2}}\right) \\
&= \text{Tr}(\hat{\mathbf{P}}_{k-1}) - \frac{\mathbf{h}_k^\top \hat{\mathbf{P}}_{k-1}^2 \mathbf{h}_k}{\mathbf{h}_k^\top \hat{\mathbf{P}}_{k-1} \mathbf{h}_k + \frac{R_k}{\beta^2}}.
\end{aligned} \tag{A10}$$

APPENDIX B: PROOF OF THEOREM 1

Here we give the proof of Theorem 1 in the article, which indicates that the optimal encoding pattern of AIME-MI under the total-energy constraint corresponds to a scanning point.

We first note that the objective function is a ratio of two positive homogeneous functions of $\mathbf{h}_k$. Following standard results in fractional programming [61], maximizing the ratio

$$\frac{\mathbf{h}_k^\top \mathbf{P}_{k-1} \mathbf{h}_k}{\mathbf{h}_k^\top \mathbf{x} + \epsilon} \tag{B1}$$

is equivalent to finding the largest scalar α such that

$$\mathbf{h}_k^\top \mathbf{P}_{k-1} \mathbf{h}_k - \alpha \mathbf{h}_k^\top \mathbf{x} - \alpha \epsilon \geq 0 \tag{B2}$$

admits a feasible solution of $\mathbf{h}_k$ under the same constraints. This leads to an equivalent parametric optimization problem as

$$\max_{\mathbf{h}_k} (\mathbf{h}_k^\top \mathbf{P}_{k-1} \mathbf{h}_k - \alpha \mathbf{h}_k^\top \mathbf{x} - \alpha \epsilon), \quad \text{s.t. } \|\mathbf{h}_k\|_1 = C, \quad \mathbf{h}_k \geq \mathbf{0}, \tag{B3}$$

where

$$\alpha = \max_{\mathbf{h}_k} \frac{\mathbf{h}_k^\top \mathbf{P}_{k-1} \mathbf{h}_k}{\mathbf{h}_k^\top \mathbf{x} + \epsilon}. \tag{B4}$$

This can be further rewritten to a standard quadratic form of $\mathbf{h}_k$:

$$\begin{aligned}
\mathbf{h}_k^\top \mathbf{P}_{k-1} \mathbf{h}_k - \alpha \mathbf{h}_k^\top \mathbf{x} - \alpha \epsilon &= \left(\mathbf{h}_k - \frac{\alpha}{2} \mathbf{P}_{k-1}^{-1} \mathbf{x}\right)^\top \mathbf{P}_{k-1} \left(\mathbf{h}_k - \frac{\alpha}{2} \mathbf{P}_{k-1}^{-1} \mathbf{x}\right) \\
&\quad - \frac{\alpha^2}{4} \mathbf{x}^\top \mathbf{P}_{k-1}^{-1} \mathbf{x} - \alpha \epsilon.
\end{aligned} \tag{B5}$$

Since $\mathbf{P}_{k-1}$ is positive definite, the objective is convex with respect to $\mathbf{h}_k$. Under the constraints $\|\mathbf{h}_k\|_1 = C$ and $\mathbf{h}_k \geq \mathbf{0}$, the feasible region is a simplex, making the extreme points correspond to scaled standard basis vectors $\{C\mathbf{e}_i\}$. Therefore, the maximum of the objective function shown as Eq. (B5) is attained at one of the extreme points, implying that the optimal solution $\hat{\mathbf{h}}_k$ corresponds to a standard basis vector.

Substituting $\mathbf{h}_k = C\mathbf{e}_i$ into the original objective yields

$$\frac{\mathbf{h}_k^\top \mathbf{P}_{k-1} \mathbf{h}_k}{\mathbf{h}_k^\top \mathbf{x} + \epsilon} = \frac{C^2 \mathbf{e}_i^\top \mathbf{P}_{k-1} \mathbf{e}_i}{C \mathbf{e}_i^\top \mathbf{x} + \epsilon} = \frac{(\mathbf{P}_{k-1})_{ii}}{x_i + \epsilon/C}. \tag{B6}$$

From this, the optimal index $\hat{i}$ can be obtained by

$$\hat{i} = \arg \max_i \frac{(\mathbf{P}_{k-1})_{ii}}{x_i + \epsilon/C}, \tag{B7}$$

which leads to the optimal $\hat{\mathbf{h}}_k = C\mathbf{e}_{\hat{i}}$ and completes the proof.

APPENDIX C: DEFINITION OF THE DETECTION SIGNAL-TO-NOISE RATIO

Here we would like to discuss the definition of the detection SNR in GI systems. In the field of GI, the bucket detection SNR (BSNR) [33] is calculated as

$$\text{BSNR} = 10 \log_{10} \left(\frac{\bar{z}^2 - \bar{z}^2}{\sigma^2} \right) \tag{C1}$$

to measure the level of signal fluctuations. However, this definition only applies to GI systems with randomly fluctuating light fields, and it is not directly related to the detection process.

In the field of signal processing, the SNR is defined as

$$\text{SNR} = 10 \log_{10} \left(\frac{P_s}{P_n} \right) = 10 \log_{10} \left(\frac{|z|^2}{|n|^2} \right), \tag{C2}$$

where P_s denotes the signal power and P_n is the noise power.

Different from traditional signal processing, which deals with electronic signals, in GI the signal is in fact an optical one. Since the detector directly measures the intensity of light fields, the recorded intensity signal can already be considered as the power. Hence, in this study we use the definition as [62]

$$\text{SNR} = 10 \log_{10} \left(\frac{\bar{z}}{\sigma} \right), \tag{C3}$$

where $\bar{z}$ represents the average value of the detection signal and σ denotes the average noise level.

APPENDIX D: DETAILED IMAGING RESULTS OVER A BROAD SAMPLING-RATIO RANGE

To complement the quantitative PSNR and SSIM curves shown in Fig. 5 of the main text, we provide the corresponding experimental reconstruction results at multiple SRs in Fig. 8. In particular, the inclusion of the 5%, 7.5%, 10%, and 12.5% cases explicitly illustrates the reconstruction performance in the low-SR regime, rather than only showing representative results at moderate or high SRs. As shown in Fig. 8, the fixed encoding baselines suffer from severe blurring or loss of structural details under low-SR conditions, whereas the AIME-based methods preserve clearer object contours and local details. These visual results are consistent with the quantitative PSNR/SSIM trends in Fig. 5 and further demonstrate the advantage of AIME-based encoding under undersampled experimental conditions.

APPENDIX E: INFLUENCE OF MODEL MISMATCH

As discussed in the main text, the AIME framework relies on a specified forward model and noise statistics for Bayesian

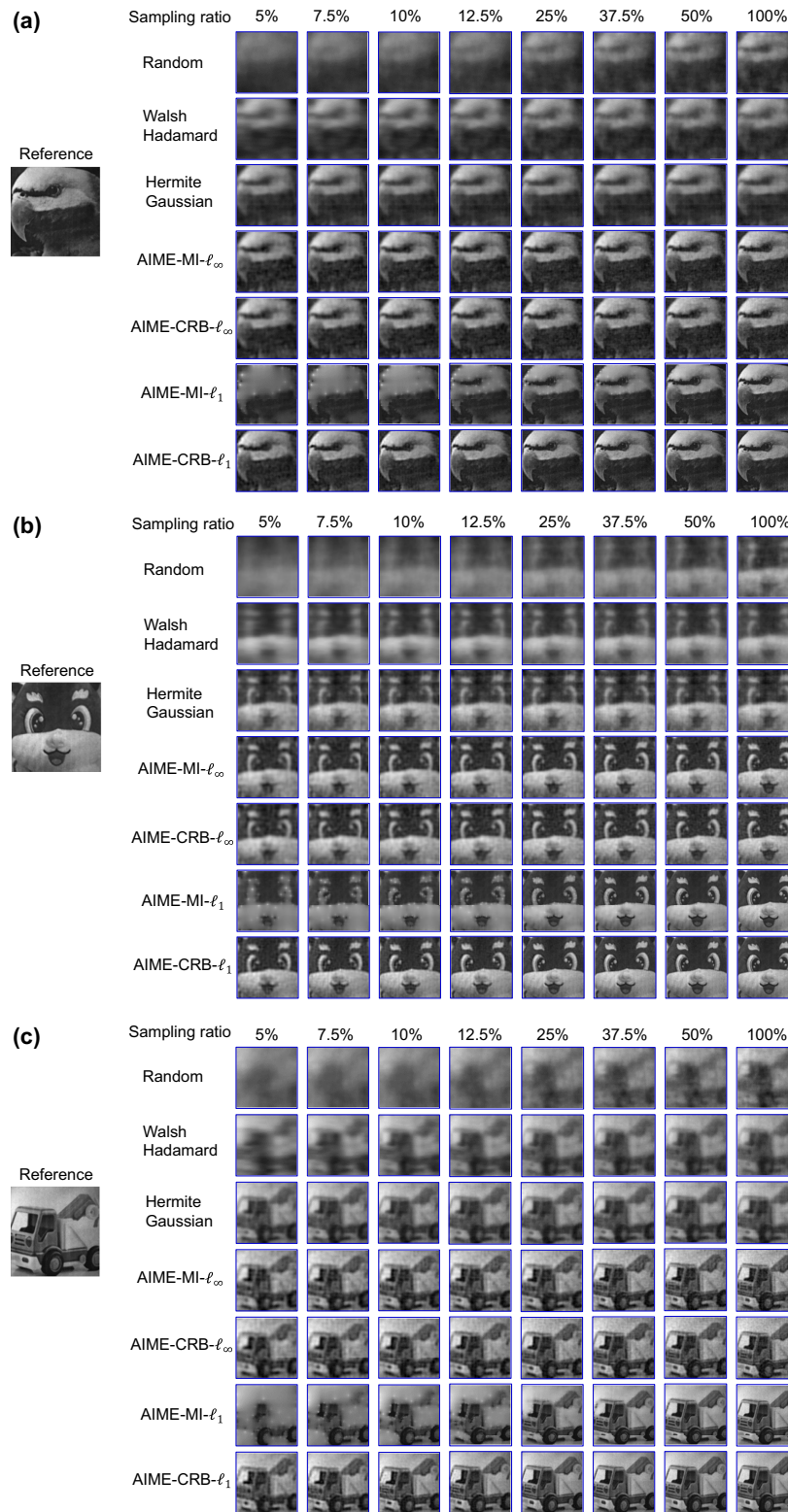


Fig. 8. Visual comparison of experimental reconstruction results under different SRs corresponding to Fig. 5. Three representative scenes are shown in (a)–(c), corresponding to the three scenes analyzed in Fig. 5 of the main text. For each scene, the reference image is shown on the left, and the reconstructed images obtained with different encoding strategies are arranged in rows. The columns correspond to SRs of 5%, 7.5%, 10%, 12.5%, 25%, 37.5%, 50%, and 100%, respectively.

inference and adaptive pattern optimization. In practice, however, the specified model may deviate from the actual measurement process because of imperfect noise calibration, system-response calibration errors, or prior mismatch. To examine the influence of such model mismatch, we performed additional robustness simulations with 128×128 images under the conditions that $\text{SNR} = 25$ dB and $\text{SR} = 25\%$. Two types of mismatch were considered: noise-statistics mismatch, where the noise parameter $\hat{\omega}$ used in Bayesian posterior updating and AIME pattern optimization differs from the ω used for generating the simulated measurements, characterized by $\alpha = \hat{\omega}/\omega$; and forward-model calibration mismatch, where the forward calibration factor $\hat{\beta}$ used in reconstruction and AIME design deviates from the β used in the simulated detection process, characterized by $\gamma = \hat{\beta}/\beta$. For both cases, $\alpha, \gamma = \{0.25, 0.5, 1, 2, 4\}$. The results corresponding to two types of mismatches are shown in Figs. 9 and 10, respectively. For noise-statistics mismatch, overestimating the noise level, corresponding to large α , makes the posterior update overly conservative and leads to blurred reconstructions, whereas underestimating the noise level makes the algorithm overconfident and introduces stronger noise-like artifacts. For calibration mismatch, an opposite trend is observed. A small γ leads to blurrier reconstructions, whereas a large γ can introduce stronger noise-like artifacts. This behavior can be understood from the effective measurement weight in the linear-Gaussian update, which depends approximately on the $\hat{\beta}^2/\hat{R}_k$. Therefore, increase on the assumed calibration factor has a

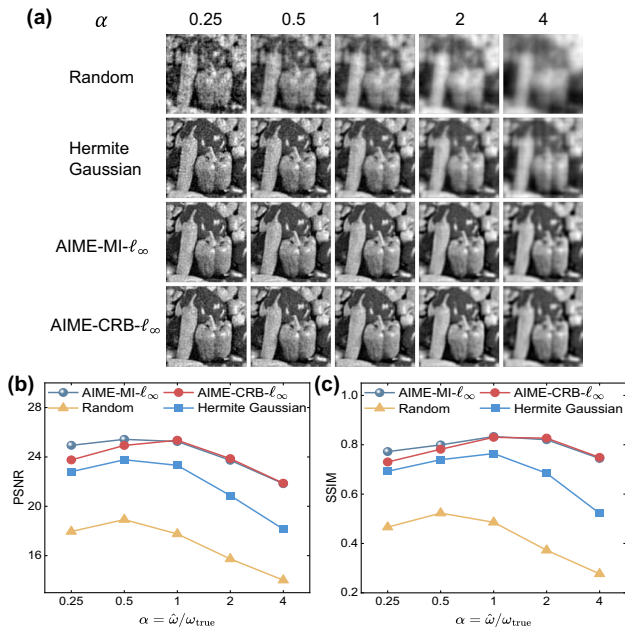


Fig. 9. Simulation results under noise-statistics mismatch. The sampling ratio is 25% and the SNR is 25 dB. The measurements were generated using the true noise parameter ω , whereas Bayesian posterior updating and AIME pattern optimization were performed using the assumed noise parameter $\hat{\omega}$. The mismatch level is defined as $\alpha = \hat{\omega}/\omega$, with $\alpha = \{0.25, 0.5, 1, 2, 4\}$. (a) Reconstructed images obtained by different encoding strategies under mismatched noise statistics. (b), (c) Quantitative comparison of PSNR and SSIM under different mismatch levels.

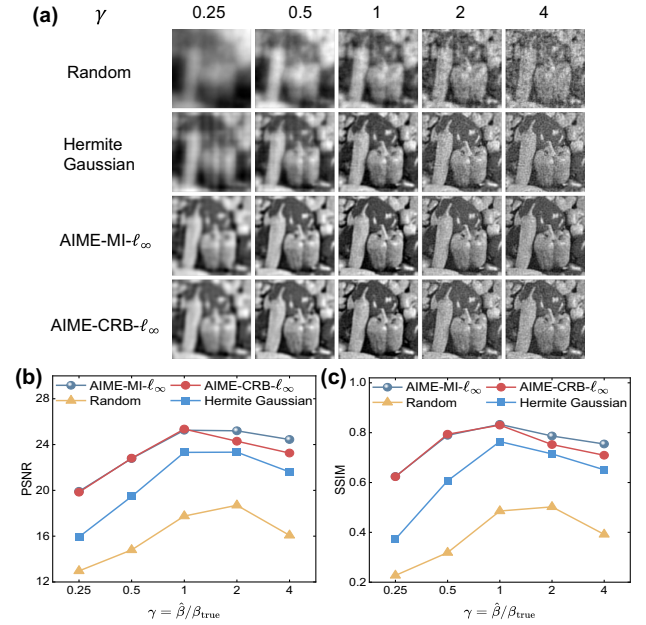


Fig. 10. Simulation results under forward-model calibration mismatch. The sampling ratio is 25% and the SNR is 25 dB. The measurements were generated using the true calibration factor β , whereas Bayesian posterior updating and AIME pattern optimization were performed using the assumed calibration factor $\hat{\beta}$. The mismatch level is defined as $\gamma = \hat{\beta}/\beta$, with $\gamma = \{0.25, 0.5, 1, 2, 4\}$. (a) Reconstructed images obtained by different encoding strategies under mismatched forward-model calibration factor. (b), (c) Quantitative comparison of PSNR and SSIM under different mismatch levels.

qualitatively similar effect to decrease on the assumed noise level. Importantly, although severe mismatch degrades the reconstruction quality, the AIME-based strategies generally retain higher structural fidelity and better quantitative performance than the fixed encoding baselines under the same mismatched Bayesian filtering framework.

The two AIME criteria also exhibit different sensitivities to mismatch. AIME-CRB is more sensitive than AIME-MI in the regime of small α or large γ . This is consistent with the CRB-based objective, in which the term R_k/β^2 explicitly controls the balance between the posterior covariance term and the assumed measurement noise. In contrast, the MI-based encoding objective is less directly affected by a global rescaling of the parameters and is mainly influenced through the posterior update. When α is large or γ is small, the assumed effective measurement SNR is low and the update becomes conservative for both criteria, so the difference between AIME-MI and AIME-CRB becomes less pronounced.

Overall, these results clarify both the robustness and the limitation of the current AIME implementation. Moderate model mismatch degrades the reconstruction quality but does not fully eliminate the advantage of AIME over the representative fixed encoding. However, severe mismatch makes the posterior update and the subsequent adaptive pattern design less reliable, thereby reducing the performance gain of AIME. These results confirm that accurate calibration of the noise statistics and system gain is important for reliable deployment.

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